PROJECT COLLECTION

REZA BABAPIRI

Control Systems & Automation

Feedback control, power converters and state estimation

4 selected documented projects

Project context / Methods / Evidence / Limitations

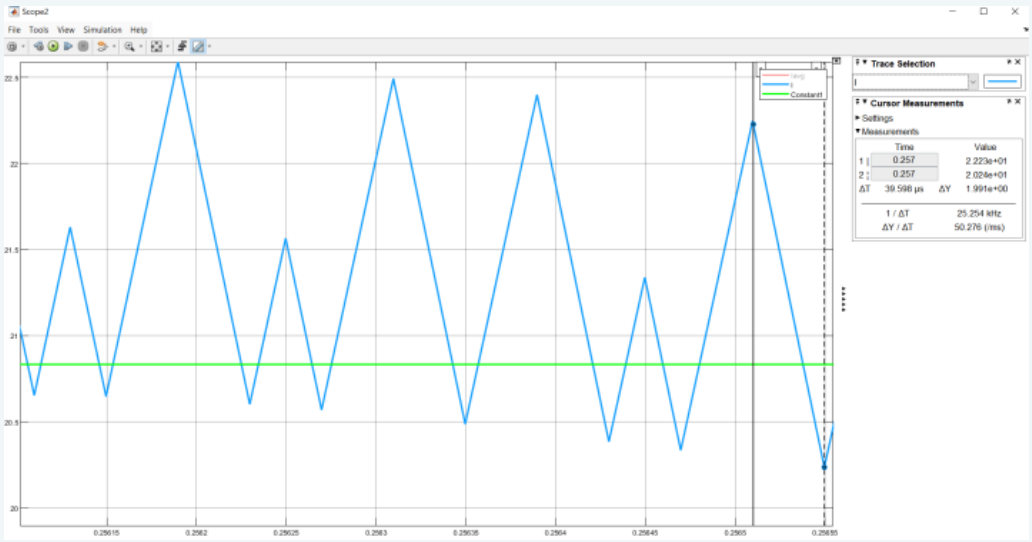
reza-babapiri.github.io

Academic work and anonymised professional experience | Source-derived summaries

POWER ELECTRONICS · SIMULINK

Buck Converter | Cascaded PI Control

100 V to 48 V, 1 kW converter design with current and voltage feedback loops



ENGINEERING SUMMARY

Designed and documented a closed-loop DC–DC buck converter using an inner current loop and outer voltage loop, with an additional single-loop model for comparison. Component sizing and controller tuning were studied through steady-state, step-response and frequency-domain simulations.

METHOD / CONTRIBUTIONS

- Sized the inductor and output capacitor around ripple requirements; compared single- and cascaded-loop structures.
- Used MATLAB scripts and Simulink models to inspect gain crossover, phase margin, transient response and load variations.
- Refined LC component values based on observed simulation behavior rather than relying on nominal sizing alone.

RESULTS & EVIDENCE

- Nominal: 100 V input, 48 V output, 1 kW, 50 kHz switching.
- Reported steady-state: 1.1% output-voltage ripple and 9.6% current ripple.
- Full-load voltage step overshoot: 1.69%; no-load case: approximately 75% overshoot (important limitation).

SCOPE / LIMITATION

No-load transient response is significantly underdamped in the supplied report. These are reported simulation results, not physical bench-test measurements.

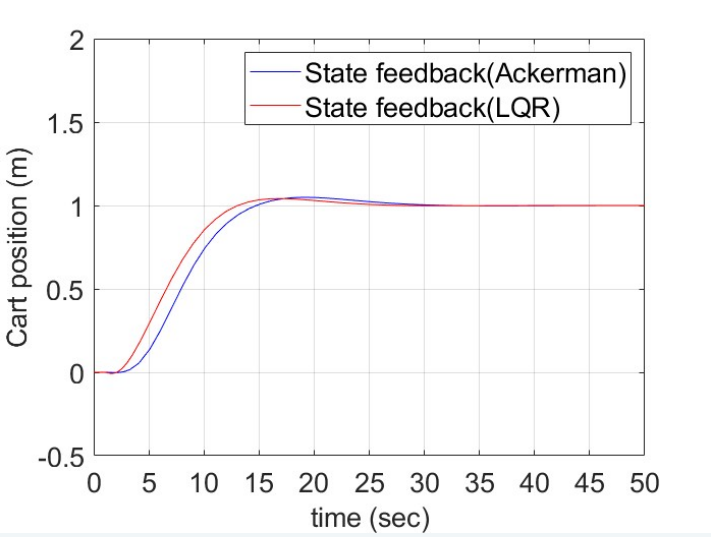
SUPPORTING RECORD

6-page report; single- and dual-loop Simulink models; MATLAB controller scripts; supporting component datasheets.

STATE-SPACE CONTROL · NONLINEAR SIMULATION

Inverted Pendulum | LQR & LQI

Cart–pendulum stabilization, tracking, saturation and anti-windup



ENGINEERING SUMMARY

Analysed the upright equilibrium of a cart–pendulum system and compared state-feedback pole placement with linear-quadratic regulator designs. Extended the study to integral tracking, actuator force constraints and nonlinear response.

METHOD / CONTRIBUTIONS

- Derived and tested a state-space model; assessed reachability and open-loop instability.
- Compared Ackermann pole placement against LQR based on transients and control effort.
- Examined integral-action reference tracking, nonlinear attraction limits, saturation and anti-windup.

RESULTS & EVIDENCE

- Demonstrated controller comparisons in report plots and MATLAB scripts.
- Identified control-effort versus transient-response trade-offs and local validity of the linear model.

SCOPE / LIMITATION

Academic simulations; the study does not demonstrate a physical stabilized pendulum.

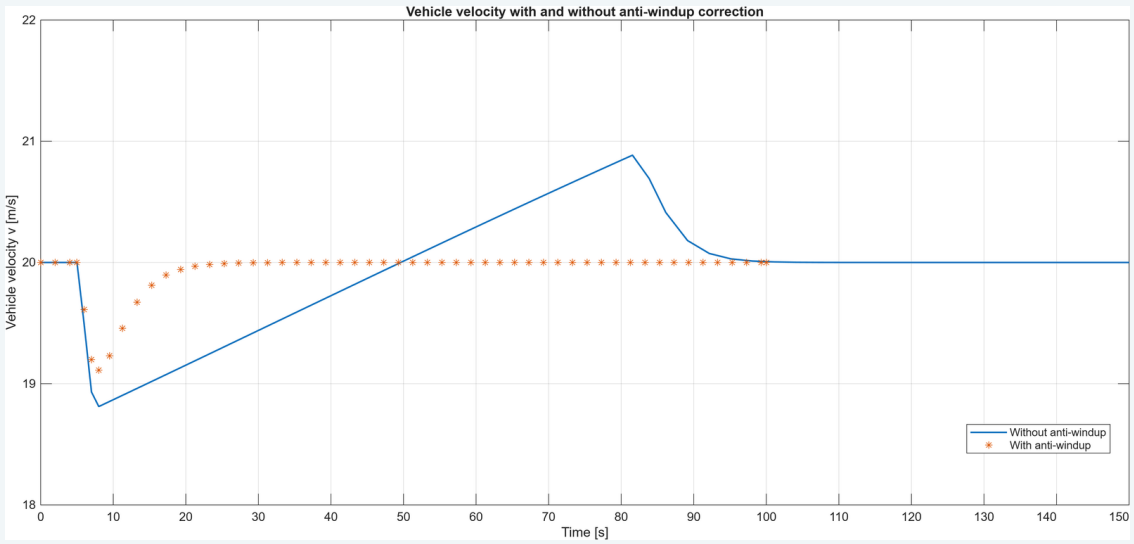
SUPPORTING RECORD

26-page CMS HW6 report and MATLAB implementation.

CONTROL ENGINEERING · ROBUSTNESS

Vehicle Cruise Control | PI & Anti-Windup

Speed tracking under hills, actuator saturation and parameter changes



ENGINEERING SUMMARY

Modelled a vehicle longitudinal plant, designed PI speed control, and explored road-grade disturbances, actuation limits, anti-windup and sensitivity functions.

METHOD / CONTRIBUTIONS

- Implemented the nonlinear vehicle model and PI control simulations.
- Introduced a road-grade disturbance and compared saturated controller behavior with and without anti-windup.
- Assessed parameter sensitivity and gang-of-four frequency-domain plots.

RESULTS & EVIDENCE

- Report comparison shows ~0.02 m/s peak overshoot with anti-windup versus ~0.885 m/s without it for one disturbance case.

SCOPE / LIMITATION

Results are specific to simulated parameters and disturbance setup.

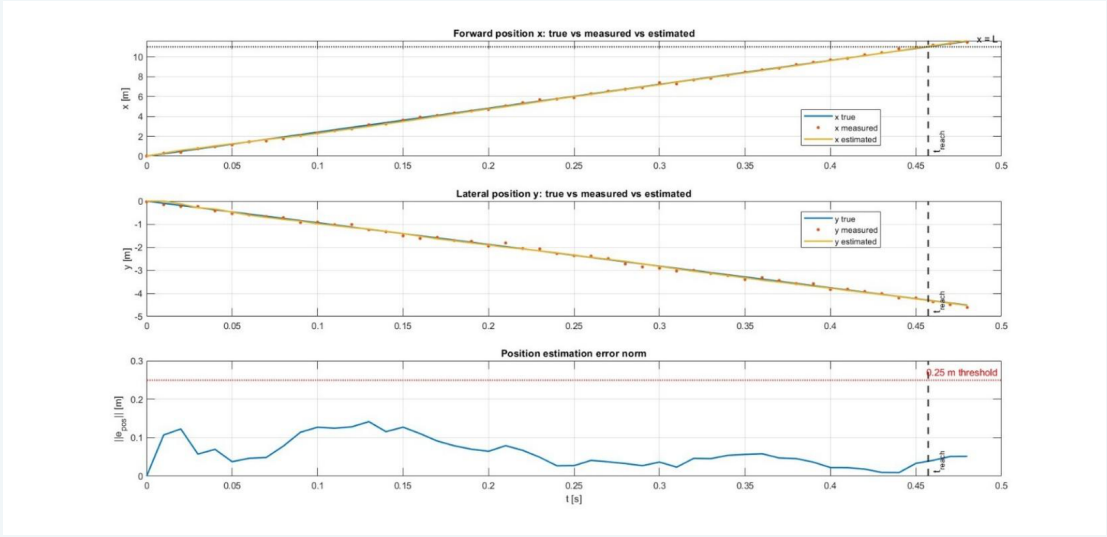
SUPPORTING RECORD

37-page CMS HW4 part 1 report; MATLAB file, Simulink models and output figures.

ESTIMATION · STOCHASTIC SYSTEMS

Kalman Filtering | Penalty-Kick Tracking

Tracking a moving target from noisy 2-D measurements



ENGINEERING SUMMARY

Built a constant-velocity state-estimation model for a football penalty kick, then studied how a Kalman filter reduces position error under noisy measurements and uncertain initial velocities.

METHOD / CONTRIBUTIONS

- Formulated a 2-D motion state and noisy measurement model.
- Simulated ten variations of the kick initial state and velocity.
- Evaluated estimation error at the target-reaching instant.

RESULTS & EVIDENCE

- The supplied report states all ten simulated cases met its 0.25 m error criterion.

SCOPE / LIMITATION

Simulation-only academic task; not a real camera-based ball tracking system.

SUPPORTING RECORD

24-page CMS HW8 report with MATLAB code and simulation figures.